

MH2100 Calculus III

Midterm II (Solutions)

AY 2021/2022, Semester 1

Quantitative Research Society @NTU

October 2021

Overview

- **Historical paper:** Midterm II, October 2021; **90 minutes** were allowed.
- **Original format:** three questions worth 50 marks in total. One two-sided A4 piece of handwritten notes was permitted.
- **Current administration (AY2026/2027):** Tuesday 6 October 2026, 16:30–18:20, venue TBA. One double-sided typed A4 sheet is permitted; no attachments or sticky notes are permitted.
- **2026 scope note:** the current midterm scope ends at double integrals, including polar coordinates. General Jacobian substitutions, cylindrical coordinates, and line integrals are beyond that scope. Question 2(v) is retained as an in-scope volume calculation by a double integral of vertical height.
- **Source:** MH2100 Midterm II, AY2021/2022 Semester 1. The questions and instructions are transcribed from the official paper; solutions and marking schemes are unofficial QRS@NTU material.

Instructions to candidates (verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. One two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(16 Marks)****QUESTION 1.**

- (i) [5 marks] Find the *maximum* and *minimum* values of $4x + 2y$ on the half disk given by $x^2 + y^2 \leq 16$ and $y \geq 0$. Also give the points (x, y) where the *maximum* and *minimum* values are achieved.

(The set of points (x, y) where $x^2 + y^2 \leq 16$ and $y \geq 0$ is a closed and bounded set so both the maximum and minimum must be achieved.)

- (ii) [3 marks] Evaluate the iterated integral.

$$\int_{-1}^1 \int_0^1 xy \, dx \, dy.$$

- (iii) [4 marks] By reversing the order of integration, evaluate the integral

$$\int_0^{30} \int_{y/6}^5 e^{x^2} \, dx \, dy.$$

- (iv) [4 marks] By reversing the order of integration, evaluate the integral

$$\int_0^4 \int_{y/4}^{\sqrt[3]{y/4}} \sin(2x^2 - x^4) \, dx \, dy.$$

Solution**Method 1: Direct computation**

- (i) Let $D = \{(x, y) : x^2 + y^2 \leq 16, y \geq 0\}$. Since D is compact and $4x + 2y$ is continuous, both extrema exist. In the interior, the gradient $(4, 2)$ does not vanish, so neither extremum is interior.

On the semicircular arc, use Lagrange multipliers for $g = x^2 + y^2 = 16$:

$$(4, 2) = \lambda(2x, 2y).$$

Thus (x, y) is in the direction $(4, 2)$. The point on the radius-4 circle is

$$(x, y) = 4 \frac{(4, 2)}{\sqrt{4^2 + 2^2}} = \left(\frac{8}{\sqrt{5}}, \frac{4}{\sqrt{5}} \right),$$

which lies on the upper arc. Its value is $8\sqrt{5}$. On the diameter $y = 0$, $-4 \leq x \leq 4$, the function is $4x$, so its endpoint values are -16 and 16 . Comparing all candidates,

$\max_D(4x + 2y) = 8\sqrt{5} \text{ at } \left(\frac{8}{\sqrt{5}}, \frac{4}{\sqrt{5}} \right), \quad \min_D(4x + 2y) = -16 \text{ at } (-4, 0).$

(ii) Integrate first with respect to x :

$$\int_{-1}^1 \int_0^1 xy \, dx \, dy = \int_{-1}^1 \left[\frac{x^2 y}{2} \right]_{x=0}^{x=1} dy = \frac{1}{2} \int_{-1}^1 y \, dy = 0.$$

Equivalently, the outer integrand is odd on $[-1, 1]$. Hence

$$\boxed{0.}$$

(iii) The original region is

$$0 \leq y \leq 30, \quad \frac{y}{6} \leq x \leq 5.$$

It is equivalently $0 \leq x \leq 5$ and $0 \leq y \leq 6x$. Reversing the order gives

$$\int_0^{30} \int_{y/6}^5 e^{x^2} \, dx \, dy = \int_0^5 \int_0^{6x} e^{x^2} \, dy \, dx = \int_0^5 6xe^{x^2} \, dx.$$

With $u = x^2$, $du = 2x \, dx$, so

$$\int_0^5 6xe^{x^2} \, dx = 3 \left[e^{x^2} \right]_0^5.$$

Therefore

$$\boxed{3(e^{25} - 1).}$$

(iv) The bounds $y/4 \leq x \leq \sqrt[3]{y/4}$ are equivalent, for $0 \leq y \leq 4$, to

$$4x^3 \leq y \leq 4x, \quad 0 \leq x \leq 1.$$

Thus, after reversing the order,

$$\begin{aligned} \int_0^4 \int_{y/4}^{\sqrt[3]{y/4}} \sin(2x^2 - x^4) \, dx \, dy &= \int_0^1 \int_{4x^3}^{4x} \sin(2x^2 - x^4) \, dy \, dx \\ &= \int_0^1 4x(1 - x^2) \sin(2x^2 - x^4) \, dx. \end{aligned}$$

Let $u = 2x^2 - x^4$. Then $du = 4x(1 - x^2) \, dx$, while $u(0) = 0$ and $u(1) = 1$. Hence

$$\int_0^1 \sin u \, du = [-\cos u]_0^1.$$

Therefore

$$\boxed{1 - \cos 1.}$$

Marking scheme (16 marks)

(i) (5 marks)

- 1 mark: uses the compactness/continuity remark or otherwise identifies that extrema are attained.
- 2 marks: correctly finds the semicircular candidate by Lagrange multipliers or an equivalent argument.
- 1 mark: checks the diameter endpoints.

- 1 mark: gives both extremal values and their points.

(ii) **(3 marks)**

- 1 mark: evaluates the inner integral correctly.
- 1 mark: evaluates the outer integral or uses oddness correctly.
- 1 mark: obtains 0.

(iii) **(4 marks)**

- 2 marks: gives the reversed bounds $0 \leq x \leq 5$, $0 \leq y \leq 6x$.
- 1 mark: integrates the y variable and makes the correct substitution.
- 1 mark: obtains $3(e^{25} - 1)$.

(iv) **(4 marks)**

- 2 marks: gives the reversed bounds $0 \leq x \leq 1$, $4x^3 \leq y \leq 4x$.
- 1 mark: obtains $4x(1 - x^2) \sin(2x^2 - x^4)$ and substitutes correctly.
- 1 mark: obtains $1 - \cos 1$.

Question 2**(17 Marks)****QUESTION 2.**

- (i) [3 marks] Find the area of the region bounded by the lines $y = x^2$ and $y = 2x + 3$.
- (ii) (*beyond 2026 midterm scope*) [2 marks] Find the Jacobian $\frac{\partial(x, y, z)}{\partial(u, v, w)}$ using the equations $x = -2u + 5$, $y = 5v + 4$, and $z = 3w - 4$.
- (iii) (*beyond 2026 midterm scope*) [4 marks] Use the transformation $u = x + y$ and $v = -2x + y$ to evaluate the integral

$$\iint_R 3x \, dA,$$

where R is the parallelogram bounded by the lines

$$y = -x + 1, \quad y = -x + 4, \quad y = 2x + 2 \quad \text{and} \quad y = 2x + 5.$$

- (iv) (*beyond 2026 midterm scope*) [4 marks] Use cylindrical coordinates to evaluate

$$\iiint_E x^2 y \, dV,$$

where E is the solid that lies between the cylinders $x^2 + y^2 = 2$ and $x^2 + y^2 = 4$, above the xy -plane and below the plane $z = y$.

- (v) [4 marks] Find the volume of the region bounded above by the sphere

$$x^2 + y^2 + z^2 = 9$$

and below by the cone $z = \sqrt{x^2 + y^2}$.

Solution**Method 1: Direct computation**

- (i) The intersections satisfy

$$x^2 = 2x + 3, \quad (x - 3)(x + 1) = 0,$$

so $-1 \leq x \leq 3$. The line lies above the parabola on this interval. Thus

$$A = \int_{-1}^3 [(2x + 3) - x^2] \, dx = \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3.$$

Therefore

$$\boxed{A = \frac{32}{3}}.$$

(ii) *Beyond 2026 midterm scope.* The Jacobian matrix is diagonal:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} -2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{vmatrix} = -30.$$

Hence

$$\boxed{\frac{\partial(x, y, z)}{\partial(u, v, w)} = -30.}$$

(iii) *Beyond 2026 midterm scope.* Solving the transformation gives

$$x = \frac{u-v}{3}, \quad y = \frac{2u+v}{3}, \quad \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{3}.$$

The two lines $y = -x + 1$ and $y = -x + 4$ become $u = 1$ and $u = 4$; the two lines $y = 2x + 2$ and $y = 2x + 5$ become $v = 2$ and $v = 5$. Thus the transformed region is the rectangle

$$1 \leq u \leq 4, \quad 2 \leq v \leq 5.$$

Moreover, $3x = u - v$. Therefore

$$\iint_R 3x \, dA = \int_1^4 \int_2^5 \frac{u-v}{3} \, dv \, du = -3.$$

Hence

$$\boxed{-3.}$$

(iv) *Beyond 2026 midterm scope.* Put $x = r \cos \theta$, $y = r \sin \theta$. Because the solid is above $z = 0$ and below $z = y$, it exists only where $y \geq 0$. Its bounds are

$$\sqrt{2} \leq r \leq 2, \quad 0 \leq \theta \leq \pi, \quad 0 \leq z \leq r \sin \theta.$$

Since $x^2 y = r^3 \cos^2 \theta \sin \theta$ and $dV = r \, dz \, dr \, d\theta$,

$$\begin{aligned} \iiint_E x^2 y \, dV &= \int_0^\pi \int_{\sqrt{2}}^2 \int_0^{r \sin \theta} r^4 \cos^2 \theta \sin \theta \, dz \, dr \, d\theta \\ &= \left(\int_0^\pi \cos^2 \theta \sin^2 \theta \, d\theta \right) \left(\int_{\sqrt{2}}^2 r^5 \, dr \right). \end{aligned}$$

The first integral is $\pi/8$, and the second is

$$\left[\frac{r^6}{6} \right]_{\sqrt{2}}^2 = \frac{64-8}{6} = \frac{28}{3}.$$

Consequently

$$\boxed{\iiint_E x^2 y \, dV = \frac{7\pi}{6}.}$$

(v) The sphere is $z = \sqrt{9 - r^2}$ above the xy -plane and the cone is $z = r$. Their intersection satisfies

$$r = \sqrt{9 - r^2}, \quad r = \frac{3}{\sqrt{2}}.$$

This volume is an in-scope double integral of the vertical height:

$$V = \int_0^{2\pi} \int_0^{3/\sqrt{2}} (\sqrt{9-r^2} - r) r \, dr \, d\theta.$$

Now

$$\int_0^{3/\sqrt{2}} \sqrt{9-r^2} r \, dr = \left[-\frac{(9-r^2)^{3/2}}{3} \right]_0^{3/\sqrt{2}} = 9 - \frac{9}{2\sqrt{2}},$$

and

$$\int_0^{3/\sqrt{2}} r^2 \, dr = \left[\frac{r^3}{3} \right]_0^{3/\sqrt{2}} = \frac{9}{2\sqrt{2}}.$$

Hence

$$V = 2\pi \left(9 - \frac{9}{2\sqrt{2}} - \frac{9}{2\sqrt{2}} \right),$$

so

$$\boxed{V = 9\pi(2 - \sqrt{2}).}$$

Marking scheme (17 marks)

- (i) **(3 marks)** 1 mark: finds both intersections; 1 mark: supplies the correct area integral; 1 mark: obtains $32/3$.
- (ii) **(2 marks)** 1 mark: forms the Jacobian matrix; 1 mark: obtains -30 .
- (iii) **(4 marks)**
 - 1 mark: solves for x, y and finds the absolute Jacobian $1/3$.
 - 2 marks: maps R to $1 \leq u \leq 4$, $2 \leq v \leq 5$ and transforms the integrand.
 - 1 mark: evaluates the integral as -3 .
- (iv) **(4 marks)**
 - 1 mark: gives correct cylindrical bounds, including $0 \leq \theta \leq \pi$.
 - 1 mark: transforms the integrand and volume element correctly.
 - 1 mark: integrates correctly.
 - 1 mark: obtains $7\pi/6$.
- (v) **(4 marks)**
 - 1 mark: finds the intersection radius $3/\sqrt{2}$.
 - 2 marks: sets up and evaluates the polar double integral of vertical height.
 - 1 mark: obtains $9\pi(2 - \sqrt{2})$.

Question 3**(17 Marks)***(beyond 2026 midterm scope)***QUESTION 3.**

- (i)
- (beyond 2026 midterm scope)*
- [4 marks] Evaluate the line integral

$$\int_C 3y \, ds,$$

where C is the part of the curve $y = x^3/3$ where $0 \leq x \leq 2$.

- (ii)
- (beyond 2026 midterm scope)*
- [4 marks] Find the work done by the force field
- $\mathbf{F}(x, y) = (y - x)\mathbf{i} + x^2\mathbf{j}$
- on a particle that moves along the curve given by
- $x = 3t$
- ,
- $y = t^2 - 1$
- from
- $(0, -1)$
- to
- $(6, 3)$
- .

- (iii)
- (beyond 2026 midterm scope)*
- [3 marks] Evaluate the line integral

$$\int_C 8x^2yz \, dx + 5z \, dy - 4xy \, dz,$$

where C has parametrisation $x = t$, $y = t^2$ and $z = t^3$, going from the point $(0, 0, 0)$ to $(1, 1, 1)$.

- (iv)
- (beyond 2026 midterm scope)*
- Let
- $\mathbf{F}(x, y, z) = (4z^2)\mathbf{i} + (\cos z)\mathbf{j} + (8xz - y \sin z + 1)\mathbf{k}$
- .

- (1) [3 marks] Show that $\mathbf{F}$ is conservative by finding a potential function f such that $\mathbf{F} = \nabla f$.
- (2) [3 marks] Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ starting at the point with $t = 0$ and ending at the point with $t = \pi$.

Solution**Method 1: Direct computation**

- (i)
- Beyond 2026 midterm scope.*
- Parametrise by
- x
- , so

$$\mathbf{r}(x) = \left\langle x, \frac{x^3}{3} \right\rangle, \quad 0 \leq x \leq 2.$$

Then

$$\left\| \frac{d\mathbf{r}}{dx} \right\| = \sqrt{1 + x^4}, \quad 3y = x^3.$$

Therefore

$$\int_C 3y \, ds = \int_0^2 x^3 \sqrt{1 + x^4} \, dx.$$

With $u = 1 + x^4$, $du = 4x^3 \, dx$, this is

$$\frac{1}{4} \int_1^{17} u^{1/2} \, du = \frac{1}{6} [u^{3/2}]_1^{17}.$$

Thus

$$\boxed{\int_C 3y \, ds = \frac{17\sqrt{17} - 1}{6}.$$

(ii) *Beyond 2026 midterm scope.* The endpoints correspond to $0 \leq t \leq 2$. Along the curve,

$$\mathbf{r}(t) = \langle 3t, t^2 - 1 \rangle, \quad \mathbf{r}'(t) = \langle 3, 2t \rangle,$$

and

$$\mathbf{F}(\mathbf{r}(t)) = \langle t^2 - 1 - 3t, 9t^2 \rangle.$$

Hence the work is

$$W = \int_0^2 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_0^2 (18t^3 + 3t^2 - 9t - 3) dt.$$

Therefore

$$W = \left[\frac{9}{2}t^4 + t^3 - \frac{9}{2}t^2 - 3t \right]_0^2, \quad \boxed{W = 56.}$$

(iii) *Beyond 2026 midterm scope.* Here

$$x = t, \quad y = t^2, \quad z = t^3, \quad 0 \leq t \leq 1,$$

so $dx = dt$, $dy = 2t dt$, and $dz = 3t^2 dt$. Substitution gives

$$\int_C 8x^2yz dx + 5z dy - 4xy dz = \int_0^1 (8t^7 + 10t^4 - 12t^5) dt.$$

Thus

$$\int_0^1 (8t^7 + 10t^4 - 12t^5) dt = 1 + 2 - 2,$$

and hence

$$\boxed{1.}$$

(iv) *Beyond 2026 midterm scope.*

(1) Integrating $f_x = 4z^2$ with respect to x gives

$$f(x, y, z) = 4xz^2 + h(y, z).$$

Since $f_y = \cos z$, we obtain $h(y, z) = y \cos z + k(z)$. Finally,

$$f_z = 8xz - y \sin z + k'(z) = 8xz - y \sin z + 1,$$

so $k'(z) = 1$. One suitable potential is

$$\boxed{f(x, y, z) = 4xz^2 + y \cos z + z.}$$

(2) The initial point is $\mathbf{r}(0) = (1, 0, 0)$ and the final point is $\mathbf{r}(\pi) = (-1, 0, \pi)$. Since $\mathbf{F} = \nabla f$,

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(-1, 0, \pi) - f(1, 0, 0) = (-4\pi^2 + \pi) - 0.$$

Therefore

$$\boxed{\int_C \mathbf{F} \cdot d\mathbf{r} = \pi - 4\pi^2.}$$

Marking scheme (17 marks)

- (i) **(4 marks)** 1 mark: parametrises C ; 1 mark: obtains $ds = \sqrt{1+x^4} dx$; 1 mark: makes the correct substitution; 1 mark: obtains $(17\sqrt{17} - 1)/6$.
- (ii) **(4 marks)** 1 mark: supplies the parameter interval and derivatives; 2 marks: forms $\mathbf{F} \cdot \mathbf{r}'$ correctly; 1 mark: obtains 56.
- (iii) **(3 marks)** 1 mark: obtains dx, dy, dz ; 1 mark: substitutes correctly; 1 mark: obtains 1.
- (iv) **(6 marks)**
 - (1) **(3 marks)** 1 mark: integrates f_x ; 1 mark: determines the y dependence; 1 mark: determines the z dependence and supplies a valid potential.
 - (2) **(3 marks)** 1 mark: finds the endpoints; 1 mark: applies the potential-difference theorem; 1 mark: obtains $\pi - 4\pi^2$.